

Solution SET: MFML (REGULAR)

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1 Question 1: Systems of Equations & Matrix Rank

(a) Solve the system $AX = b$

Problem:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & -1 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 6 \\ 12 \\ 5 \end{bmatrix}$$

Method 1: Standard Gaussian Elimination

Step 1: Form the Augmented Matrix $[A|b]$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & 4 & 6 & 12 \\ 1 & -1 & 0 & 5 \end{array} \right]$$

Step 2: Elimination (Targeting R_2 and R_3)

- Perform $R_2 \rightarrow R_2 - 2R_1$:

$$\begin{bmatrix} 2 & 4 & 6 & 12 \end{bmatrix} - 2 \begin{bmatrix} 1 & 2 & 3 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

- Perform $R_3 \rightarrow R_3 - R_1$:

$$\begin{bmatrix} 1 & -1 & 0 & 5 \end{bmatrix} - \begin{bmatrix} 1 & 2 & 3 & 6 \end{bmatrix} = \begin{bmatrix} 0 & -3 & -3 & -1 \end{bmatrix}$$

Resulting Matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 0 & 0 & 0 \\ 0 & -3 & -3 & -1 \end{array} \right]$$

Step 3: Organize to Echelon Form Swap R_2 and R_3 to put the zero row at the bottom:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -3 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Step 4: Back Substitution

- From R_3 : $0 = 0$ implies infinite solutions. Let $z = t$ (free variable).
- From R_2 : $-3y - 3z = -1 \implies -3y = 3t - 1 \implies y = \frac{1}{3} - t$.
- From R_1 : $x + 2y + 3z = 6$

$$x + 2\left(\frac{1}{3} - t\right) + 3t = 6 \implies x + \frac{2}{3} + t = 6 \implies x = \frac{16}{3} - t$$

Solution: $X = \begin{bmatrix} 16/3 \\ 1/3 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$

Method 2: Intuitive Row Observation (The Fast Way)

Observation: Look closely at Row 1 and Row 2.

- $R_1 : 1x + 2y + 3z = 6$
- $R_2 : 2x + 4y + 6z = 12$

Notice that $R_2 = 2 \times R_1$. This means the second plane is identical to the first. It adds no new constraints. We can discard it immediately.

Simplified System: We are left with only two equations:

1. $x + 2y + 3z = 6$
2. $x - y = 5$ (From original R_3)

Solving: From (2), we get $x = y + 5$. Substitute this directly into (1):

$$(y + 5) + 2y + 3z = 6 \implies 3y + 3z = 1 \implies y = \frac{1}{3} - z$$

Let $z = t$, then $y = 1/3 - t$ and $x = (1/3 - t) + 5 = 16/3 - t$.

(b) Echelon Form, Determinant, Rank of B

$$B = \begin{bmatrix} 5 & 25 & -64 & 32 \\ 0 & -1 & -3 & 4 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 5 & -3 \end{bmatrix}$$

Method 1: Row Echelon Form Calculation

Step 1: Eliminate the 1 in R_3 using R_2 .

$$R_3 \rightarrow R_3 + R_2 : \quad [0 \quad 1 \quad -2 \quad 5] + [0 \quad -1 \quad -3 \quad 4] = [0 \quad 0 \quad -5 \quad 9]$$

New Matrix:

$$\begin{bmatrix} 5 & 25 & -64 & 32 \\ 0 & -1 & -3 & 4 \\ 0 & 0 & -5 & 9 \\ 0 & 0 & 5 & -3 \end{bmatrix}$$

Step 2: Eliminate the 5 in R_4 using R_3 .

$$R_4 \rightarrow R_4 + R_3 : \quad [0 \quad 0 \quad 5 \quad -3] + [0 \quad 0 \quad -5 \quad 9] = [0 \quad 0 \quad 0 \quad 6]$$

Final Echelon Form:

$$U = \begin{bmatrix} 5 & 25 & -64 & 32 \\ 0 & -1 & -3 & 4 \\ 0 & 0 & -5 & 9 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

Determinant: $\det(B) = 5 \times (-1) \times (-5) \times 6 = 150$.

Rank: 4 (Four non-zero rows).

Method 2: Laplace Expansion (Block Method)

Expand along **Column 1** because it has three zeros $[5, 0, 0, 0]^T$.

$$\det(B) = 5 \times \det \begin{bmatrix} -1 & -3 & 4 \\ 1 & -2 & 5 \\ 0 & 5 & -3 \end{bmatrix}$$

Now expand this 3×3 minor along its **Column 1** $[-1, 1, 0]^T$:

$$\begin{aligned} &= 5 \left[(-1) \det \begin{bmatrix} -2 & 5 \\ 5 & -3 \end{bmatrix} - (1) \det \begin{bmatrix} -3 & 4 \\ 5 & -3 \end{bmatrix} \right] \\ &= 5[-1(6 - 25) - 1(9 - 20)] \\ &= 5[-1(-19) - 1(-11)] = 5[19 + 11] = 5(30) = 150 \end{aligned}$$

2 Question 2: Vector Spaces

(a) Linear Dependence

Vectors: $v_1 = [5, -3, 1]^T$, $v_2 = [6, 2, -3]^T$, $v_3 = [4, -8, 5]^T$.

Method 1: Solving the Linear System

We want to find scalars c_1, c_2 such that $c_1 v_1 + c_2 v_2 = v_3$. Augmented Matrix:

$$\left[\begin{array}{cc|c} 5 & 6 & 4 \\ -3 & 2 & -8 \\ 1 & -3 & 5 \end{array} \right]$$

Solving: 1. From R_3 : $c_1 - 3c_2 = 5 \implies c_1 = 3c_2 + 5$. 2. Substitute into R_2 :

$$-3(3c_2 + 5) + 2c_2 = -8 \implies -9c_2 - 15 + 2c_2 = -8 \implies -7c_2 = 7 \implies c_2 = -1$$

3. Find c_1 : $c_1 = 3(-1) + 5 = 2$. 4. Verify with R_1 : $5(2) + 6(-1) = 10 - 6 = 4$. (Correct).

Conclusion: Since $v_3 = 2v_1 - v_2$, the vectors are Linearly Dependent.

Method 2: Determinant Test

Three vectors in $\mathbb{R}^3$ are dependent if the volume of the parallelepiped they form is zero.

$$\det(A) = \det \begin{bmatrix} 5 & 6 & 4 \\ -3 & 2 & -8 \\ 1 & -3 & 5 \end{bmatrix}$$

Expansion along Row 1:

$$\begin{aligned} & 5(10 - 24) - 6(-15 - (-8)) + 4(9 - 2) \\ &= 5(-14) - 6(-7) + 4(7) \\ &= -70 + 42 + 28 = 0 \end{aligned}$$

Conclusion: Determinant is zero, so vectors are Linearly Dependent.

(b) Prove that V is a subspace; Find Basis & Dimension

Method 1: Formal Subspace Proof

To prove: $V = \text{span}\{v_1, v_2\}$ is a subspace of $\mathbb{R}^3$.

- Zero Vector:** Let coefficients be 0. $0v_1 + 0v_2 = \mathbf{0}$. Thus $\mathbf{0} \in V$.
- Closure under Addition:** Let $u, w \in V$. $u = a_1v_1 + a_2v_2$ and $w = b_1v_1 + b_2v_2$.
 $u + w = (a_1 + b_1)v_1 + (a_2 + b_2)v_2$. This is still a linear combination of v_1, v_2 , so $u + w \in V$.
- Closure under Scalar Multiplication:** Let $k \in \mathbb{R}$. $ku = k(a_1v_1 + a_2v_2) = (ka_1)v_1 + (ka_2)v_2$. This is still a linear combination, so $ku \in V$.

Basis: Since v_1 is not a scalar multiple of v_2 , they are Linearly Independent. Basis = $\{[5, -3, 1]^T, [6, 2, -3]^T\}$. **Dimension:** 2.

Method 2: Span Property (The Intuitive Way)

Theorem: The span of any set of vectors in a vector space is always a subspace of that vector space. Since V is defined explicitly as $\text{span}\{v_1, v_2\}$, it is by definition a subspace.

Dimension Check: The matrix $[v_1 v_2]$ has rank 2 (columns are independent). Therefore, Dimension = 2.

(c) Check if $u = [4, -8, 2]^T$ is in V

Method 1: Standard Consistency Check

Try to solve $c_1 v_1 + c_2 v_2 = u$:

$$\left[\begin{array}{cc|c} 5 & 6 & 4 \\ -3 & 2 & -8 \\ 1 & -3 & 2 \end{array} \right]$$

From R_3 : $c_1 - 3c_2 = 2 \implies c_1 = 3c_2 + 2$. Substitute into R_2 : $-3(3c_2 + 2) + 2c_2 = -8 \implies -7c_2 = -2 \implies c_2 = 2/7$. Check R_1 : $5(3(2/7) + 2) + 6(2/7) = 5(20/7) + 12/7 = 112/7 = 16$. But RHS is 4. Since $16 \neq 4$, the system is **Inconsistent**.

Method 2: Normal Vector Strategy (Geometric)

1. Find the normal vector n to the plane spanned by v_1, v_2 .

$$n = v_1 \times v_2 = \det \begin{bmatrix} i & j & k \\ 5 & -3 & 1 \\ 6 & 2 & -3 \end{bmatrix}$$

$$= i(9 - 2) - j(-15 - 6) + k(10 + 18) = [7, 21, 28]^T$$

Simplify direction: $n = [1, 3, 4]^T$. 2. The plane equation is $1x + 3y + 4z = 0$. 3. Check point $u = [4, -8, 2]^T$:

$$1(4) + 3(-8) + 4(2) = 4 - 24 + 8 = -12 \neq 0$$

Conclusion: The point does not satisfy the plane equation, so it is not in V .

3 Question 3: Singular Value Decomposition (Detailed)

(a) Find the SVD of $A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$

Approach 1: The Standard Algorithm ($A^T A$)

Step 1: Compute Gram Matrix $A^T A$

$$A^T A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

Step 2: Find Singular Values (Σ) Solve characteristic equation for $A^T A$:

$$\det(A^T A - \lambda I) = (2 - \lambda)^2 - 4 = \lambda^2 - 4\lambda = 0$$

Eigenvalues: $\lambda_1 = 4, \quad \lambda_2 = 0$.

Singular Values ($\sigma = \sqrt{\lambda}$): $\sigma_1 = 2, \quad \sigma_2 = 0$.

$$\Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

Step 3: Find Right Singular Vectors (V) Solve $(A^T A - \lambda I)x = 0$ for each λ .

- For $\lambda = 4$: $\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \implies x_1 = x_2$. Unit vector: $v_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$.
- For $\lambda = 0$: $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \implies x_1 = -x_2$. Unit vector: $v_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$.

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Step 4: Find Left Singular Vectors (U)

- Use formula $u_1 = \frac{1}{\sigma_1} A v_1$:

$$u_1 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2/\sqrt{2} \\ -2/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

- For u_2 , since $\sigma_2 = 0$, find a vector orthogonal to u_1 :

$$u_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Approach 2: Independent Diagonalization (AA^T)

Use this method to double-check U , or if calculation of Av_1 is difficult.

Step 1: Compute AA^T

$$AA^T = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$$

Step 2: Find Eigenvectors of AA^T

- **For $\lambda = 4$:** $\begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix} x = 0 \implies x_1 = -x_2$. Vector: $\begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$. (Matches u_1).
- **For $\lambda = 0$:** $\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} x = 0 \implies x_1 = x_2$. Vector: $\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$. (Matches u_2).

Conclusion: This confirms the U matrix derived in Approach 1 is correct.

(b) Is the matrix A diagonalizable?

Method 1: AM vs GM Comparison

1. **Eigenvalues of A :** Trace=0, Det=0. Characteristic eq: $\lambda^2 = 0$. Roots: $\lambda = 0, 0$. **Algebraic Multiplicity (AM) = 2.**
2. **Geometric Multiplicity (GM):** We find the nullity of $(A - 0I)$.

$$A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

The rank is 1. By Rank-Nullity Theorem:

$$GM = \text{Nullity} = n - \text{Rank} = 2 - 1 = 1$$

3. **Conclusion:** Since $GM(1) \neq AM(2)$, the matrix is **NOT diagonalizable**.

Method 2: Nilpotent Property (Conceptual)

Compute A^2 :

$$A^2 = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Since A is a non-zero nilpotent matrix ($A \neq 0$ but $A^k = 0$), it cannot be diagonalized. (The only diagonalizable nilpotent matrix is the zero matrix itself).

4 Question 4: Multivariate Calculus

Function: $f(x, y) = x^2 + xy + 2y^2 - 4x + 3$.

(a) Compute the gradient $\nabla f(x, y)$

Solution

Compute partial derivatives:

$$\frac{\partial f}{\partial x} = 2x + y - 4$$

$$\frac{\partial f}{\partial y} = x + 4y$$

Gradient Vector:

$$\nabla f = \begin{bmatrix} 2x + y - 4 \\ x + 4y \end{bmatrix}$$

(b) Find the critical point(s)

Method: Solving the System

Set $\nabla f = \mathbf{0}$:

$$1. \ x + 4y = 0 \implies x = -4y$$

$$2. \ 2x + y - 4 = 0$$

Substitute (1) into (2):

$$2(-4y) + y - 4 = 0 \implies -8y + y = 4 \implies -7y = 4 \implies y = -4/7$$

Find x :

$$x = -4(-4/7) = 16/7$$

Critical Point: $(16/7, -4/7)$

(c) Classify the critical point

Hessian Matrix $H = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$.

Method 1: Sylvester's Criterion (Principal Minors)

We calculate the leading principal minors D_k :

$$1. \ D_1 = H_{11} = 2.$$

$$2. \ D_2 = \det(H) = (2)(4) - (1)(1) = 7.$$

Analysis: Since $D_1 > 0$ and $D_2 > 0$, the matrix is **Positive Definite**. A positive definite Hessian at a critical point implies a **Local Minimum**.

Method 2: Eigenvalue Analysis (General Method)

A symmetric matrix is Positive Definite if and only if all its eigenvalues are positive. Solve $\det(H - \lambda I) = 0$:

$$\begin{aligned}(2 - \lambda)(4 - \lambda) - 1 &= 0 \\ \lambda^2 - 6\lambda + 8 - 1 &= \lambda^2 - 6\lambda + 7 = 0\end{aligned}$$

Using quadratic formula:

$$\lambda = \frac{6 \pm \sqrt{36 - 28}}{2} = \frac{6 \pm \sqrt{8}}{2} = 3 \pm \sqrt{2}$$

Values:

$$\lambda_1 \approx 3 + 1.414 = 4.414$$

$$\lambda_2 \approx 3 - 1.414 = 1.586$$

Analysis: Since both eigenvalues are strictly positive (> 0), the function curves upward in all directions. **Result:** Local Minimum.

(d) Find Taylor series of $g(x) = f(x, 1)$ at $x = 1$

Method 1: Calculus (Derivatives)

1. Define $g(x)$ by setting $y = 1$ in $f(x, y)$:

$$g(x) = x^2 + x(1) + 2(1)^2 - 4x + 3 = x^2 - 3x + 5$$

2. Compute values at center $x = 1$:

$$g(1) = 1 - 3 + 5 = 3$$

$$g'(x) = 2x - 3 \implies g'(1) = -1$$

$$g''(x) = 2 \implies g''(1) = 2$$

3. Taylor Formula: $g(x) = g(1) + g'(1)(x - 1) + \frac{g''(1)}{2!}(x - 1)^2$

$$g(x) = 3 - (x - 1) + (x - 1)^2$$

Method 2: Algebraic Substitution (Intuitive)

Let $x - 1 = h \implies x = 1 + h$. Substitute into $g(x) = x^2 - 3x + 5$:

$$g(1 + h) = (1 + h)^2 - 3(1 + h) + 5$$

$$= (1 + 2h + h^2) - 3 - 3h + 5$$

$$= h^2 - h + 3$$

Replace h with $(x - 1)$:

$$g(x) = 3 - (x - 1) + (x - 1)^2$$

5 Question 5: Inner Product Spaces

(a) Properties of Inner Product & Proof

Matrix: $A = \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix}$. Define $\langle X, Y \rangle = X^T A Y$.

Method 1: Standard Verification

For this to be a valid inner product on $\mathbb{R}^2$, A must be:

1. **Symmetric:** Is $A = A^T$? Yes, $A_{12} = A_{21} = -1$. **Proof:** $\langle Y, X \rangle = Y^T A X = (Y^T A X)^T$ (scalar transpose is itself) $= X^T A^T Y = X^T A Y = \langle X, Y \rangle$.
2. **Positive Definite:** Is $X^T A X > 0$ for $X \neq 0$? Using Sylvester's Criterion: $D_1 = 3 > 0$. $D_2 = \det(A) = 3 - 1 = 2 > 0$. Since all principal minors are positive, A is positive definite.

Method 2: Completing the Square (Alternative Proof)

Expand the quadratic form $X^T A X$ for $X = [x_1, x_2]^T$:

$$Q(x) = 3x_1^2 + x_2^2 - 2x_1x_2$$

Rearrange to show it is always positive:

$$\begin{aligned} &= 2x_1^2 + (x_1^2 - 2x_1x_2 + x_2^2) \\ &= 2x_1^2 + (x_1 - x_2)^2 \end{aligned}$$

Since squared terms are non-negative, and both cannot be zero unless $x_1 = x_2 = 0$, the form is Positive Definite.

(b) Find the distance between vectors

Vectors $u = [1, -1]^T$ and $v = [2, 1]^T$.

Method 1: Matrix Multiplication

1. Find difference vector $w = u - v = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$. 2. Distance squared is $\langle w, w \rangle_A = w^T A w$.

$$A w = \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ -2 \end{bmatrix} = \begin{bmatrix} -3 + 2 \\ 1 - 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

3. Multiply by w^T :

$$w^T (A w) = [-1 \quad -2] \begin{bmatrix} -1 \\ -1 \end{bmatrix} = 1 + 2 = 3$$

Distance: $\sqrt{3}$.

Method 2: Algebraic Quadratic Form

Use the form from (a): $Q(x) = 3x^2 + y^2 - 2xy$. Substitute $w = [-1, -2]^T$ (so $x = -1, y = -2$):

$$\begin{aligned} &= 3(-1)^2 + (-2)^2 - 2(-1)(-2) \\ &= 3(1) + 4 - 4 \\ &= 3 \end{aligned}$$

Distance: $\sqrt{3}$.

(c) Verify Orthogonality

Vectors $x = [-1, 2]^T$ and $y = [2, 1]^T$.

Solution

Two vectors are orthogonal if their inner product is zero.

$$\langle x, y \rangle_A = x^T A y = \begin{bmatrix} -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Calculate Ay :

$$\begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 - 1 \\ -2 + 1 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \end{bmatrix}$$

Calculate Dot Product:

$$\begin{bmatrix} -1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \end{bmatrix} = -5 - 2 = -7$$

Conclusion: Since $-7 \neq 0$, the vectors are **NOT** orthogonal.